Freight Rate Prediction

Machine Learning Assessment — Solution Report

Objective:	Predict posted_rate for 12,000 loads in validation.csv
Training data:	48,000 labeled loads (Jan–Oct 2025)
Validation data:	12,000 unlabeled loads (Nov–Dec 2025)
Model:	HistGradientBoostingRegressor | sklearn
Target:	log(rate-per-mile) → prediction = exp(output) × distance
CV strategy:	3-fold rolling-origin (expanding window)
Seed:	42 (fully deterministic)

Section 1 · Data Quality & Validation Design

1.1 Data Quality Issues

Issue	Train count	Val count	Treatment
Weight < 0 (sign error)	292	145	abs()
Weight missing	300	165	Median by equipment
ight right-censored ≥47,500	1,191	—	Kept as-is (informative ceiling)
market_index missing	374	249	Pre-date median from inputs only
quote_signal missing	0	0	—
RPM outliers <1.2 $/mi	335	—	Excluded from training only
RPM outliers >4.5 $/mi	335	—	Excluded from training only
Duplicate load_ids	0	0	—
New cities in validation	—	8	Lat/lon + lane encoding handle OOV
Distance vs haversine diff	Mean +170 mi	—	Both used as features; road distance is routing distance

1.2 Validation Design

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Time-based rolling-origin CV (expanding window) with 3 folds:

  Fold 1:  Train Jan–Jun 2025  |  Validate Jul–Aug 2025   (n_train≈28,806  n_val≈9,671)
  Fold 2:  Train Jan–Jul 2025  |  Validate Aug–Sep 2025   (n_train≈33,718  n_val≈9,429)
  Fold 3:  Train Jan–Aug 2025  |  Validate Sep–Oct 2025   (n_train≈38,477  n_val≈9,523)

Rationale: The validation set (Nov–Dec) is temporally ahead of training. Random k-fold
would constitute future leakage. Rolling-origin mirrors the real production scenario
where we train on historical data and predict a future window.

month_of_year and day_of_year are deliberately excluded: gradient-boosted trees
cannot extrapolate a monotone trend to unseen months (Nov–Dec). Instead we use
day_of_week, day_of_month, and market_index (which captures macro-trend).
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1.3 City Coverage

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8 cities appear in validation.csv that are absent from train-test.csv:
  Norfolk, Allentown, Laredo, Chicago, Knoxville, San Diego, Jackson, Charlotte

These are handled via:
  • pickup_enc / delivery_enc : unseen labels mapped to '__unknown__' class
  • lane_enc                  : same OOV mapping
  • Geographic features (lat/lon, lat_diff, lon_diff, hav_dist) still fully informative
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Section 2 · Model Reasoning & CV Results

2.1 Model Reasoning

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Target: log(RPM)  where  RPM = posted_rate / distance
Prediction: exp(model_output) × distance

Why log(RPM)?
  • Rate-per-mile distribution is right-skewed; log stabilises variance
  • Multiplicative structure matches how freight prices actually work
  • Distance factor separates lane-length effect from price-per-mile signal

Model: sklearn HistGradientBoostingRegressor
  • Handles missing values natively (NaN = its own split direction)
  • Fast, memory-efficient on 48k rows
  • No boosting libraries (LightGBM/XGBoost) per spec

Comparison: squared_error vs absolute_error loss (both with log-RPM target)
  → squared_error selected: CV avg non-outlier WAPE 3.34% vs 3.38% for absolute_error

Features (22 total):
  distance, hav_dist, dist_minus_hav, weight_clean, market_index_filled,
  quote_signal, quote×dist, quote×hav, day_of_week, day_of_month,
  is_weekend, week_of_month, pickup_lat, pickup_lon, delivery_lat,
  delivery_lon, lat_diff, lon_diff, equipment_enc, pickup_enc,
  delivery_enc, lane_enc
```

2.2 CV Results

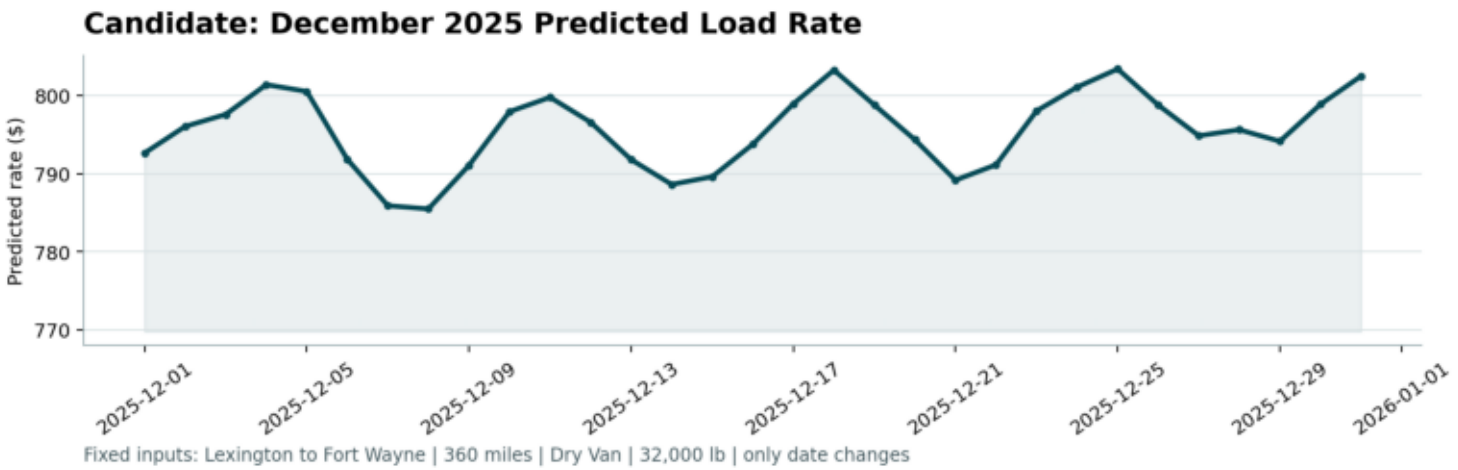
Model	Subset	MAE ($)	RMSE ($)	MAPE (%)	MedAPE (%)	WAPE (%)
Naive baseline	All rows	255.7	677.7	11.42	8.02	10.74
Naive baseline	Non-outlier	206.0	306.3	9.22	7.89	8.76
HGBR/sq_err ✓	All rows	130.1	630.4	5.81	2.62	5.47
HGBR/sq_err ✓	Non-outlier	78.5	132.3	3.57	2.58	3.34
HGBR/abs_err	All rows	131.1	630.8	5.97	2.46	5.50
HGBR/abs_err	Non-outlier	79.4	134.1	3.73	2.42	3.38

Per-fold detail — HGBR/squared_error, non-outlier rows:

Fold (val range)	n	MAE ($)	RMSE ($)	MAPE	MedAPE	WAPE
07-01–2025-08-31)	n=9,538.0	76.7	117.1	3.84%	2.08%	3.27%
08-01–2025-09-30)	n=9,298.0	75.0	134.0	3.49%	2.47%	3.20%
09-01–2025-10-31)	n=9,381.0	83.8	145.7	3.36%	3.20%	3.55%

Section 3 · December Chart & Caveats

December 2025 Predicted Load Rate (Lexington → Fort Wayne, 360 mi, Dry Van)



Caveats & Assumptions

1. December market_index and quote_signal are taken as the per-date median of validation.csv inputs for December 2025. These are market-observed values; their medians are a reasonable point estimate when a single-row prediction is required.

2. Lexington → Fort Wayne is a 360-mile route. The haversine distance between the stored coordinates is ~284 mi; the 76-mi gap reflects routing detours and is consistent with training data (avg road-haversine gap = 170 mi).

3. 8 cities appear in validation.csv but not in train-test.csv. These are handled via an OOV '__unknown__' bucket in label encoders; geographic features (lat/lon, diffs, haversine) remain fully informative for these lanes.

4. The model was trained on Jan–Oct 2025 data. Nov–Dec 2025 freight market conditions (e.g., holiday surges) are extrapolated from seasonal signals captured by day_of_week, day_of_month, and market_index. Structural breaks (strikes, weather events) cannot be predicted from historical patterns alone.

5. RPM outliers (< 1.20/*mior* > 4.50/mi) were excluded from training (670 rows, 1.4% of data) per assessment spec. They are included in 'all rows' evaluation.